

KULLIYAH OF INFORMATION & COMMUNICATION TECHNOLOGY

MOBILE APPLICATION DEVELOPMENT | CSCI 4311

SEMESTER 2, 2025/2026

SECTION 1

Title:

IjabQabul

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INTRODUCTION

Planning a wedding is an important stage in a person's life, but it can also be stressful and overwhelming. For Muslim couples in Malaysia, wedding preparation involves more than just selecting a venue, catering service, or wedding attire. Couples must also complete various religious and legal requirements, prepare supporting documents, attend compulsory courses, manage expenses, and ensure that all procedures are completed before their wedding date. Since marriage procedures differ from one state to another, many couples often struggle to find accurate information and keep track of what needs to be done.

Nowadays, smartphones have become an essential part of everyday life. People use mobile applications to manage their schedules, finances, communication, and daily tasks. However, many existing wedding planning applications are designed for a general audience and may not cater specifically to the needs of Muslim couples in Malaysia. As a result, couples often have to rely on multiple sources such as websites, social media platforms, messaging applications, and personal notes to organize their wedding preparations. This can make the planning process more complicated, especially for first-time couples who are unfamiliar with the required procedures.

The idea for IjabQabul: Wedding Planner Application was inspired by these challenges. The application is designed to serve as a centralized platform where couples can manage their wedding preparations more effectively. Instead of searching for information from different sources or manually creating their own schedules and checklists, users can access all the necessary planning tools within a single application.

Several features are proposed to support users throughout their wedding planning journey. The personalized checklist feature helps users organize and monitor important tasks, ensuring that no essential preparation is overlooked. The smart timeline feature automatically generates a recommended schedule based on the couple's wedding date, role, and state, helping them complete tasks according to a suitable timeline. In addition, the wedding package comparison feature allows users to compare available packages and services based on price, inclusions, and reviews before making decisions. The application also provides state specific marriage guidance, enabling users to understand the procedures, required documents, and legal requirements applicable to their respective states.

Flutter has been selected as the main development framework because it supports cross platform mobile development, allowing the application to run on both Android and iOS devices using a single codebase. Firebase will be used as the Backend-as-a-Service (BaaS) platform to support user authentication, cloud based data storage, and data synchronization. The combination of these technologies enables the development of a reliable, scalable, and user-friendly mobile application.

In conclusion, IjabQabul aims to simplify the wedding preparation process for Muslim couples in Malaysia by providing an organized and accessible digital solution. By bringing

together wedding planning tools, marriage guidance, and progress tracking features in one platform, the application can help users manage their preparations more efficiently, reduce stress, and make better decisions throughout their wedding journey.

1. PROJECT IDEATION & INITIATION

A. Title

IjabQabul : Wedding Planner Application

B. Background Problem

IjabQabul is a mobile application designed to help Muslim couples in Malaysia manage wedding and marriage preparations more efficiently. Wedding planning can be stressful due to multiple tasks, budgeting, package selection and different state marriage procedures.

To address these challenges, our application provides personalized checklists, collaboration and timeline features, wedding package comparison and state-specific marriage guidance in a centralized platform. IjabQabul aims to simplify wedding preparations while helping couples save time, reduce stress and make better planning decisions.

C. Purpose of Objective

1. Provide a centralized platform for preparation management.
2. Help users organize wedding tasks through personalized wedding checklists and timeline.
3. Simplify wedding package comparison and planning.
4. Provide accessible state-specific marriage guidance for Muslim couples in Malaysia.
5. Improve collaboration between couples during wedding preparation.

D. Target User

The primary target users are first-time marrying Muslim couples in Malaysia aged between 20 and 35 years old. This represents a significant market segment, with the Department of Statistics Malaysia (DOSM) Marriage and Divorce Statistics recording approximately 137,595 Muslim marriages annually out of around 200,000 total annual marriages nationwide.

E. Preferred Platform

Preferred Platform: The preferred platform for this project is Smartphones, targeting both Android and iOS operating systems through Flutter's cross-platform capabilities. For the user interface development, the application explicitly implements Material Design widgets rather than Cupertino components. This approach ensures a unified, highly consistent layout structure and seamless gesture navigation across both mobile environments while optimizing the overall user experience for smartphone screens. Wearable platforms are excluded from the current development scope to focus entirely on mobile device compatibility.

F. Features & Functionalities

The IjabQabul mobile application offers a centralized platform equipped with the following core features and functionalities designed to streamline the wedding preparation process for Muslim couples in Malaysia:

- **Personalized Checklists:** A static, categorized repository of vendor, budget, and generic logistical tasks that users can check off and track at their own pace.
- **Smart Collaboration and Timeline Features:** A dynamic, chronological roadmap structured entirely around the couple's specific wedding (*nikah*) date. Instead of forcing users to build a schedule manually, the application collects key onboarding data including the user's role (groom or bride) and their specific Malaysian state. The app then automatically generates a tailored milestone timeline with recommended deadlines, reducing confusion and automatically integrating state-specific legal procedures into the user's schedule.
- **Wedding Package Comparison:** A structured comparison utility allowing users to evaluate different wedding vendors, services, and packages based on pricing, inclusions, and reviews to support informed budgeting decisions.
- **State-Specific Marriage Guidance:** A comprehensive navigation resource providing step-by-step legal procedures, documentation requirements, and official administrative workflows tailored to the unique regulations of individual Malaysian states.

2. REQUIREMENT ANALYSIS & PLANNING

2.1 Requirement Analysis

2.1.1 Technical feasibility and Back End assessment

The proposed IjabQabul mobile application is technically feasible to build using Flutter. This is primarily due to Flutter cross platform capabilities which allow it to be implemented on both iOS and Android using essentially a single codebase. Additionally, Flutter offers a vast ecosystem of packages, a large selection of widgets, and high performance, all of which are very helpful when developing mobile applications.

The app itself focuses on managing wedding preparations rather than complex computational or real time processing. Therefore, generally this project should be feasible within the allotted development duration while still meeting the functional criteria outlined in the project scope.

2.1.2 Data storage and CRUD operations

The application needs Create, Read, Update, and Delete (CRUD) operations for handling user generated details. The suggested database structure is going to include:

Module	CRUD Operations
User Profile	Create user profile, update personal information, view profile details
Wedding Checklist	Add tasks, view tasks, edit task status, delete tasks
Wedding Timeline	Create milestones, update progress, view schedule
Package Comparison	Save package selections, update comparison data, remove saved packages
Marriage Guidance	Read only access to state specific information

Table 1: CRUD Operations

Firebase Firestore is recommended for backend implementation since it provides a cloud based NoSQL data store, secure authentication services, and real-time synchronisation that

simply keeps everything in sync. It enables flexibility for future improvements and has an easy interaction with Flutter which is quite helpful.

2.1.3 Packages and Plugins Assessment

A few Flutter packages are needed in order to support what the application is doing, basically to cover the required functionality.

Package	Purpose
firebase_core	Firestore initialization
cloud_firestore	Cloud database storage
firebase_auth	User authentication
provider	State management
table_calendar	Timeline and calendar display
shared_preferences	Local data storage
flutter_markdown	Display state guidance content
intl	Date and time formatting
flutter_slidable	Interactive task management
url_launcher	Open official state marriage portals

Table 2: Packages

These packages are stable, widely used, and compatible with the latest Flutter framework.

2.1.4 Platform Compatibility Assessment

1. Smartphone Compatibility

The application is primarily designed for smartphone platforms, such as iOS 14 and later and Android 10 and later. Smaller displays don't feel too strange thanks to Flutter's responsive layout features, which help maintain a fairly similar user experience across various screen sizes and resolutions.

2. Wearable Compatibility

Smartwatches and other wearable technology are not included in the project's scope. Mostly because the screen size is constrained and the wedding preparation processes become more intricate than they first appear.

So, in short, IjabQabul will be optimized exclusively for smartphone platforms.

2.1.5 Logical Design

Sequence Diagram: User Wedding Planning Workflow

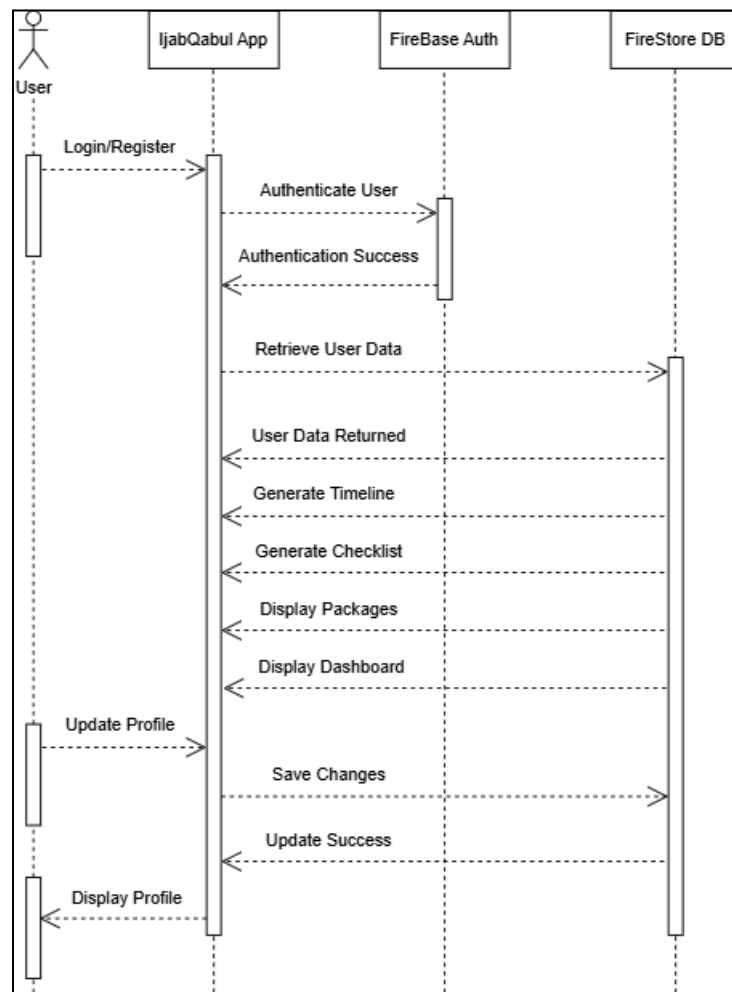


Figure 1: Sequence Diagram of IjabQabul

The diagram illustrates how the IjabQabul Wedding Planner Application interacts with its users and backend services in a way that feels seamless and supportive. When a user begins by logging in or registering, the app communicates with Firebase Authentication to verify their identity. Once authentication is successful, the app retrieves the user's data from Firestore. This data is then used to generate personalized features such as the wedding timeline, checklist, package comparisons, and the main dashboard. The flow ensures that couples immediately see relevant information tailored to their profile without needing to repeat steps or provide extra input.

Later, when the user updates their profile, the app saves the changes directly to Firestore. Once the update is confirmed, the profile is displayed with the new information. This process highlights how the system maintains consistency across all modules while keeping the experience smooth and reliable. In simple terms, the diagram shows the journey of a couple interacting with the app, from logging in to managing their wedding details, with Firebase services working quietly in the background to keep everything secure and synchronized.

Screen Navigation Flow

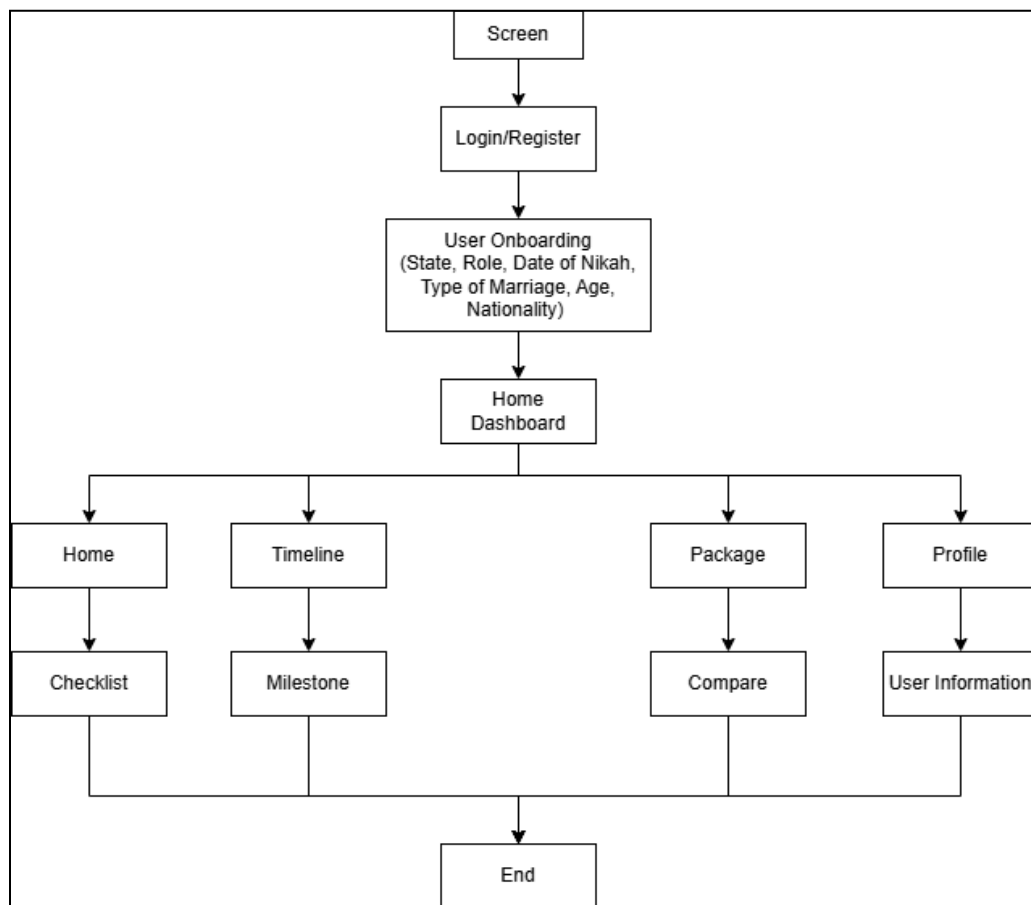


Figure 2: Screen Flow of IjabQabul

The flowchart represents the journey of a user inside the IjabQabul Wedding Planner Application in a way that feels intuitive and organized. It begins with the login or registration process, followed by onboarding where the user provides important personal details such as their state, role, date of nikah, type of marriage, age, and nationality. Once this information is captured, the user is directed to the home dashboard, which acts as the central hub for all wedding planning activities.

From the dashboard, the user can explore different sections of the app. The home section leads to the personalized checklist, the timeline section generates milestones, the package section allows comparisons of wedding packages, and the profile section stores user information. Each of these modules is connected back to the dashboard, ensuring that the experience is smooth and consistent. The flow ultimately concludes when the user finishes their tasks, showing how the app guides couples step by step through their wedding preparation while keeping everything centralized and easy to manage.

The navigation structure follows a hierarchical design that allows users to access major wedding planning functionalities directly from the main dashboard.

2.2 Planning

The development of the IjabQabul mobile application follows an Agile inspired Software Development Life Cycle (SDLC) methodology. Agile was selected because it provides flexibility, encourages continuous collaboration among team members, and allows improvements to be made throughout the development process. Unlike traditional development approaches where each phase is completed only once, Agile promotes iterative development, enabling the team to identify issues, gather feedback, and make necessary enhancements before the final release.

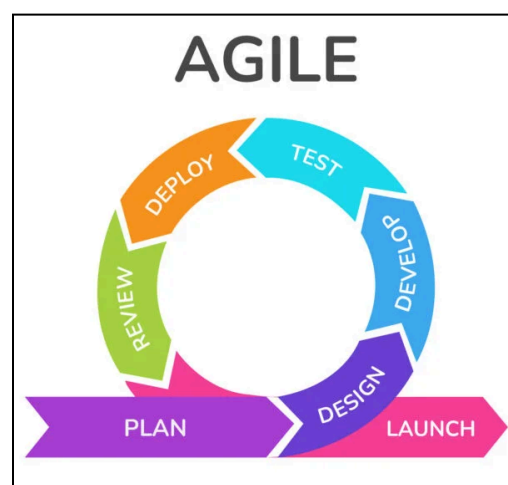


Figure 3: Agile Methodology

For this project, the Agile inspired approach is adapted into seven main phases which are Planning, Requirement Analysis, Design, Development, Testing, Deployment, Review, and Launch. These phases ensure that the application is developed systematically while maintaining flexibility to accommodate improvements throughout the project lifecycle.

2.2.1 Planning

The planning phase serves as the foundation of the entire project. During this phase, the development team identifies the main problem that the application intends to solve and establishes the overall direction of the project. Extensive discussions are conducted among team members to understand the difficulties commonly faced by Muslim couples in Malaysia during wedding preparation, including managing wedding tasks, budgeting, coordinating with vendors, and understanding state-specific marriage procedures.

The team also determines the scope of the project, identifies the target users, and establishes realistic goals that can be achieved within the project timeline. In addition, the technologies to be used throughout development, including Flutter and Firebase, are evaluated and selected based on their suitability for mobile application development.

Proper planning is crucial because it helps minimize project risks, ensures that team members share a common understanding of the project objectives, and provides a clear roadmap for subsequent development activities.

Activities:

- Identifying the problem statement
- Defining project objectives and scope
- Identifying target users
- Conducting preliminary research
- Selecting development technologies
- Planning project schedule and milestones
- Assigning responsibilities among team members

Deliverables:

- Project proposal
- Project scope document
- Initial project schedule

- Team task allocation plan

2.2.2 Requirement Analysis

The requirement analysis phase focuses on identifying, documenting, and analysing all requirements necessary for the successful implementation of the application. During this stage, the team determines what features and functionalities should be included in the system to address the identified problems effectively.

Both functional and non-functional requirements are examined. Functional requirements describe the services and features that the application must provide, such as user authentication, wedding checklist management, timeline generation, package comparison, and state specific marriage guidance. Non-functional requirements focus on aspects such as usability, performance, reliability, scalability, and security.

The team also evaluates the technical feasibility of implementing the proposed features using Flutter and Firebase. Data storage requirements, CRUD operations, package compatibility, and platform support are carefully assessed to ensure that the project can be completed within the available resources and timeframe.

This phase is important because a clear understanding of user and system requirements reduces development errors and ensures that the final product aligns with user expectations.

Activities:

- Gathering user requirements
- Identifying functional requirements
- Identifying non-functional requirements
- Evaluating technical feasibility
- Analysing CRUD operations
- Assessing database requirements
- Evaluating Flutter packages and plugins

Deliverables:

- Requirement specification document
- Functional requirement analysis
- Non-functional requirement analysis

- Technical feasibility report
- Database requirement specification

2.2.3 Design

The design phase translates the requirements gathered during the analysis stage into detailed visual and technical representations of the system. This phase focuses on planning how the application will look, how users will interact with it, and how different components of the system will communicate with each other.

The user interface (UI) is designed using Material Design principles to ensure consistency and usability across Android and iOS devices. Wireframes and mockups are created to visualize the layout of each screen before development begins. The user experience (UX) is also carefully planned to ensure that navigation remains intuitive and easy to understand, particularly for first-time users.

In addition to interface design, the team develops sequence diagrams, screen navigation flows, database structures, and system architecture diagrams. These designs serve as blueprints that guide the development process and reduce the likelihood of implementation errors.

The design phase ensures that all stakeholders have a clear understanding of how the application will function before coding begins.

Activities:

- Creating wireframes and mockups
- Designing user interfaces
- Planning user experience flows
- Developing screen navigation diagrams
- Creating sequence diagrams
- Designing database structures
- Establishing design standards and guidelines

Deliverables:

- UI wireframes
- UX flow diagrams
- Screen navigation flow

- Sequence diagrams
- Database schema
- System architecture design

2.2.4 Development

The development phase is where the actual implementation of the application takes place. Based on the approved designs and requirements, the team begins coding the application using Flutter as the development framework and Firebase as the Backend-as-a-Service platform.

Development is carried out in modules to improve manageability and facilitate testing. Each major feature of the application is developed separately before being integrated into the complete system. Core functionalities include user registration and login, personalized wedding checklists, smart timeline generation, package comparison, and state specific marriage guidance.

The team also integrates Firebase Authentication for secure user login and Firestore Database for cloud based data storage. Additional Flutter packages are incorporated to support specific functionalities such as calendar displays, state management, and local storage.

Version control is managed through GitHub, enabling team members to collaborate efficiently and maintain code quality throughout development.

Activities:

- Setting up the Flutter development environment
- Configuring Firebase services
- Developing authentication functionality
- Implementing CRUD operations
- Building application screens
- Integrating packages and plugins
- Managing source code using GitHub
- Performing continuous integration

Deliverables:

- Authentication module
- Checklist management module

- Timeline module
- Package comparison module
- Marriage guidance module
- Integrated application prototype

2.2.5 Testing

Testing is conducted continuously throughout development to ensure that all functionalities operate as intended and meet user requirements. The primary objective of this phase is to identify defects, improve application stability, and verify that all features function correctly across supported devices.

Different testing methods are applied, including functionality testing, input validation testing, user interface testing, and integration testing. The team verifies that data is stored and retrieved correctly from Firebase, user authentication functions securely, and navigation flows operate without errors.

Any issues identified during testing are documented, corrected, and retested to ensure successful resolution before deployment.

Activities:

- Functional testing
- User interface testing
- Input validation testing
- Integration testing
- Bug fixing and debugging
- Performance verification

Deliverables:

- Test reports
- Bug tracking documentation
- Improved application stability

2.2.6 Deployment

Following successful testing, the application is prepared for deployment in a controlled environment. This phase ensures that all application components are properly configured and ready for demonstration or future production use.

Firebase services are finalized, application configurations are verified, and final builds are generated. The team performs additional checks to ensure that backend services, authentication mechanisms, and data synchronization operate correctly in the deployed environment.

The deployment phase also allows the team to evaluate the application's readiness before presenting it to lecturers and evaluators.

Activities:

- Finalizing Firebase configuration
- Generating application builds
- Verifying backend connectivity
- Conducting deployment testing
- Preparing demonstration environment

Deliverables:

- Deployable application build
- Deployment documentation

2.2.7 Review and Launch

The review and launch phase represents the final stage of the project lifecycle. During this phase, the completed application is evaluated against the original project objectives, scope, and requirements. The team reviews all implemented features to ensure that the application successfully addresses the identified problems.

Feedback from lecturers, peers, and potential users is collected and analysed. Any final improvements are incorporated before the official project presentation. The application is then formally launched through a live demonstration, where the team showcases its features, technical implementation, and benefits.

This phase also includes preparing all supporting documentation, GitHub repositories, and presentation materials required for project submission.

Activities:

- Reviewing project objectives and outcomes
- Collecting stakeholder feedback
- Implementing final improvements
- Preparing project documentation
- Conducting final presentation and demonstration
- Submitting project deliverables

Deliverables:

- Final mobile application
- GitHub repository
- Project documentation
- Final presentation
- Project submission package

By following these Agile inspired phases, the development team can ensure that the IjabQabul mobile application is developed systematically, efficiently, and in accordance with the project requirements while maintaining flexibility for continuous improvement throughout the development lifecycle.

2.3 Gantt Chart and Project Timeline

Start Date: 18th May 2026

Final Presentation Date: 29th June 2026

Total Project Duration: Approximately 8 Weeks

NO.	ACTIVITY	MAY				JUNE			
		WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 1	WEEK 2	WEEK 3	WEEK 4
		4/5 - 10/5	11/5 - 17/5	18/5 - 24/5	25/5 - 31/5	1/6 - 7/6	8/6 - 14/6	15/6 - 21/6	22/6 - 29/6
1	Project Initiation								
2	Requirement Analysis								
3	System Design								
4	UI/UX Design								
5	Development : Authentication								
6	Development : Checklist								
7	Development : Timeline								
8	Development : Package Comparison								
9	Development : Marriage Guidance								
10	Testing & Debugging								
11	User Acceptance Testing								
12	Documentation								
13	Final Improvements								
14	Presentation Preparation								
15	Final Project Presentation								

Figure 4: Gantt Chart of Development IjabQabul

The development of the IjabQabul Wedding Planner Application was carried out in a structured and progressive way over eight weeks, beginning in May and ending with the final presentation in June. Each stage of the project was carefully planned and executed, starting with identifying the challenges faced by Muslim couples in Malaysia during wedding preparation and moving through requirement analysis, system design, UI/UX finalization, and feature development. Core modules such as authentication, personalized checklist, smart timeline, package comparison, marriage guidance, and user profile management were built and tested independently before being integrated into the main system.

By combining Flutter for the frontend and Firebase for the backend, the team created a responsive and reliable application that simplifies complex wedding tasks. The modular architecture improved maintainability and teamwork, while GitHub ensured smooth collaboration and version control. In the end, the application achieved its objectives by offering couples a centralized, user-friendly tool to manage their wedding preparations with confidence. Future enhancements such as AI-based budget recommendations, vendor booking integration, and real-time collaboration could make the app even more comprehensive and supportive.

3. PROJECT DESIGN

The Project Design phase is about turning system requirements into a mobile application that feels intuitive, visually appealing, and easy to use. Because the IjabQabul application is intended for first-time Muslim couples in Malaysia, the interface is created to be simple, organized, and welcoming regardless of the user's technical background. The design follows Google's Material Design guidelines, which provide a consistent visual language and responsive interactions across both Android and iOS devices developed with Flutter.

Before moving into implementation, the team prepared wireframes and high-fidelity mockups to visualize the layout, navigation flow, and interaction between modules. Each screen was carefully designed with usability in mind so that users could quickly find important information and complete wedding planning tasks without unnecessary effort.

The design work focused on three main aspects. The first was the User Interface, which ensures that the visual elements are clear and attractive. The second was the User Experience, which emphasizes smooth navigation and logical flows that guide users step by step. The third was Design Consistency, which makes sure that every part of the application feels connected and coherent, giving couples confidence as they move through their wedding planning journey.

3.1 User Interface (UI)

The User Interface of the IjabQabul application is designed with modern mobile application principles that emphasize simplicity, clarity, accessibility, and responsiveness. Because smartphones have smaller screens compared to desktop computers, every element is carefully positioned to maximize space while maintaining readability and ease of interaction.

Flutter's Material Design widgets are used throughout the application to provide a familiar and consistent experience for both Android and iOS users. Common components such as the AppBar, Bottom Navigation Bar, Cards, TextFields, ElevatedButtons, Floating Action Buttons, ListTiles, and SnackBars are implemented so that users can interact naturally without needing additional guidance.

The layout is kept clean with sufficient spacing between components to avoid clutter. Icons are paired with descriptive text labels, making it easier for users to recognize functions. Important actions such as saving data, editing information, or deleting checklist items are highlighted with distinctive button styles and colors to improve visibility.

Navigation follows a clear structure where users begin at the login screen before accessing the main dashboard. From the dashboard, they can move easily to all major modules including the Wedding Checklist, Smart Timeline, Package Comparison, Marriage Guidance, and Profile Management. This straightforward flow reduces unnecessary steps and improves efficiency.

To support mobile gestures, Flutter's built-in gesture detection is applied across the application. Users can tap buttons to perform actions, swipe checklist items to edit or delete tasks, scroll through long content, and select dates using interactive calendar widgets. These gestures follow familiar smartphone patterns, making the application intuitive even for first-time users.

Responsive layouts are achieved using Flutter widgets such as Expanded, Flexible, MediaQuery, and SingleChildScrollView. This ensures that the interface adapts smoothly to different screen sizes and resolutions, providing a consistent viewing experience across a wide range of Android and iOS devices.

3.2 User Experience (UX)

The User Experience of the IjabQabul application is designed to make wedding planning feel simple, organized, and less stressful. Wedding preparation often involves handling many tasks, deadlines, documents, and vendors, so the application reduces mental strain by presenting information in a clear and structured way.

The design follows a user-centered approach where each feature supports the natural flow of wedding preparation. After logging in, users are directed straight to the dashboard, which acts as the central hub. From this screen, they can reach every major function without having to navigate through unnecessary menus.

Each module is created to require only the essential steps needed to complete a task. For example, adding a new checklist item asks for basic information and saves it directly to Firebase. The Smart Timeline automatically generates milestones based on the wedding date and state, removing the need for manual scheduling.

Navigation remains consistent across the application. Buttons, page titles, and action icons are placed in the same locations on every screen so users always know how to return to the previous page, access the home dashboard, or update their profile.

Feedback is provided after every important action. When a task is saved, a profile updated, or a checklist item completed, confirmation messages appear to reassure the user. If incorrect or incomplete information is entered, validation messages appear immediately to guide them. This interactive feedback builds confidence and reduces mistakes.

Accessibility is also considered. Readable font sizes, strong color contrast, large touch targets, and clear icons make the application comfortable to use for people with different levels of mobile experience.

Overall, the UX design guides users naturally through each stage of wedding preparation. It minimizes frustration, avoids unnecessary complexity, and encourages couples to stay engaged with the application throughout their planning journey.

3.3 Consistency

Consistency is one of the most important design principles applied throughout the IjabQabul application. A consistent visual identity helps users become familiar with the system more quickly and reduces the learning curve when moving between different modules.

A unified colour palette is used across all screens, combining soft green, white, and gold to reflect simplicity, elegance, and the theme of an Islamic wedding. Green conveys harmony and growth, while gold represents celebration and happiness. Together these colours create a pleasant visual identity and reinforce the overall branding of the application.

Typography is also kept consistent. The same font families, text sizes, and heading styles are applied to page titles, section headings, labels, and body content. This improves readability and gives the application a professional appearance.

The IjabQabul logo appears consistently on the splash screen, login page, and application header to strengthen brand recognition. Forms follow the same design pattern with identical input fields, labels, spacing, and button styles. This makes it easier for users to complete forms because each screen behaves in a familiar way.

Interactive components such as buttons, cards, icons, and navigation bars share the same styling across the application. Primary actions use a consistent button colour and shape, while secondary actions follow similar treatments in every module. This reduces confusion because users quickly learn how different interface elements function.

Every screen follows the same overall layout structure with an AppBar at the top, content displayed in the centre, and navigation controls positioned in predictable locations. This layout helps users navigate confidently without needing to relearn the interface whenever they move to another section.

By maintaining consistency in colours, typography, icons, navigation patterns, forms, and interaction behaviours, the application delivers a cohesive user experience. This approach improves usability, enhances visual appeal, and supports efficient wedding planning for couples who rely on the system.

3.4 Mockup Design Explanation

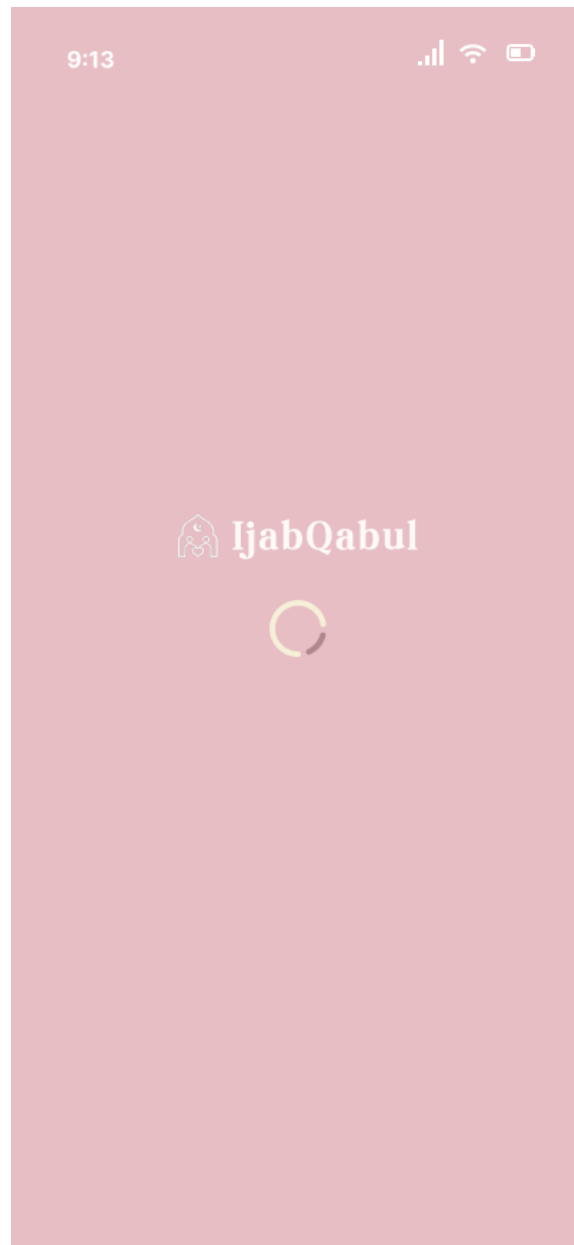


Figure 5: Splash Screen

The Splash Screen serves as the entry point of the IjabQabul application. It displays the application's logo and branding while the system initializes Firebase services and checks whether the user has an active login session. The simple design with minimal elements provides a professional first impression while reducing loading distractions before directing users to the appropriate screen.

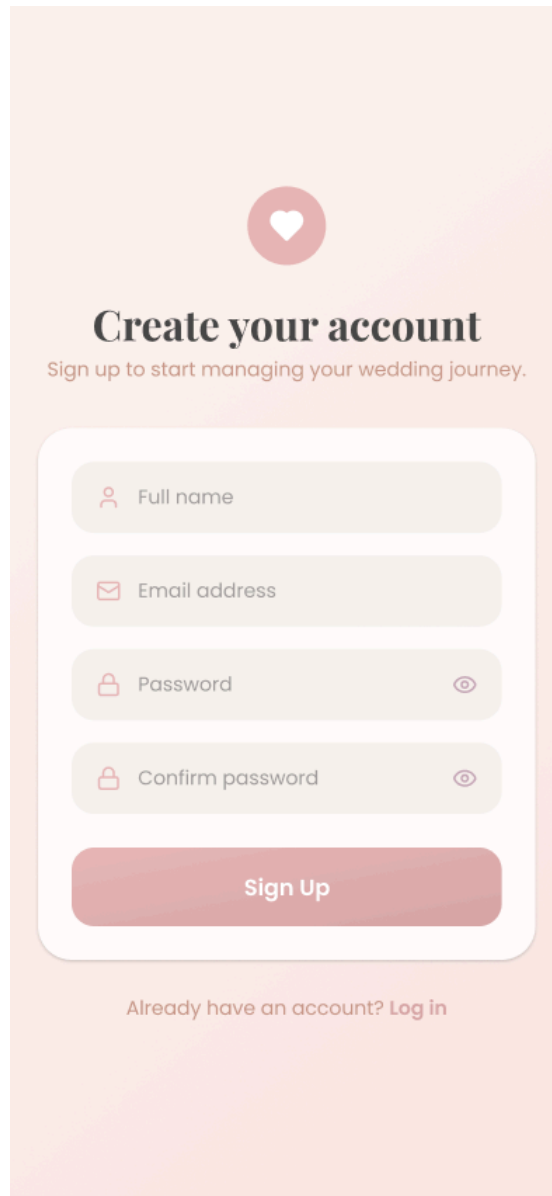
The image shows a mobile application registration screen with a light pink background. At the top center is a red circle containing a white heart icon. Below this is the title "Create your account" in bold black text, followed by the subtitle "Sign up to start managing your wedding journey." in a smaller, lighter font. The registration form is a white rounded rectangle containing four input fields: "Full name" with a person icon, "Email address" with an envelope icon, "Password" with a lock icon and a toggle eye icon, and "Confirm password" with a lock icon and a toggle eye icon. A red "Sign Up" button is at the bottom of the form. Below the form, the text "Already have an account? Log in" is displayed in a small, light font.

Figure 6: Registration Screen

The Registration Screen allows new users to create an account by entering their personal information. Input validation ensures that all required fields are completed correctly before registration. Once successful, user information is securely stored in Firebase Authentication and Cloud Firestore.

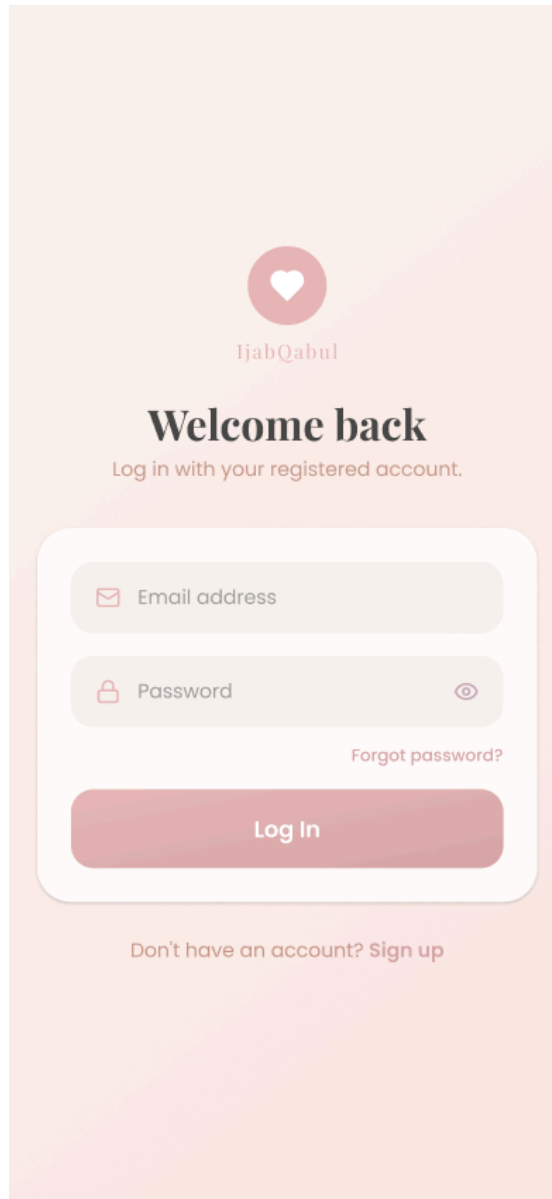


Figure 7: Login Screen

The Login Screen allows registered users to securely access their personal wedding planning information using Firebase Authentication. The interface contains clearly labelled email and password input fields, a login button, and a navigation option for new users to create an account. The clean layout minimizes distractions and enables users to complete authentication quickly.

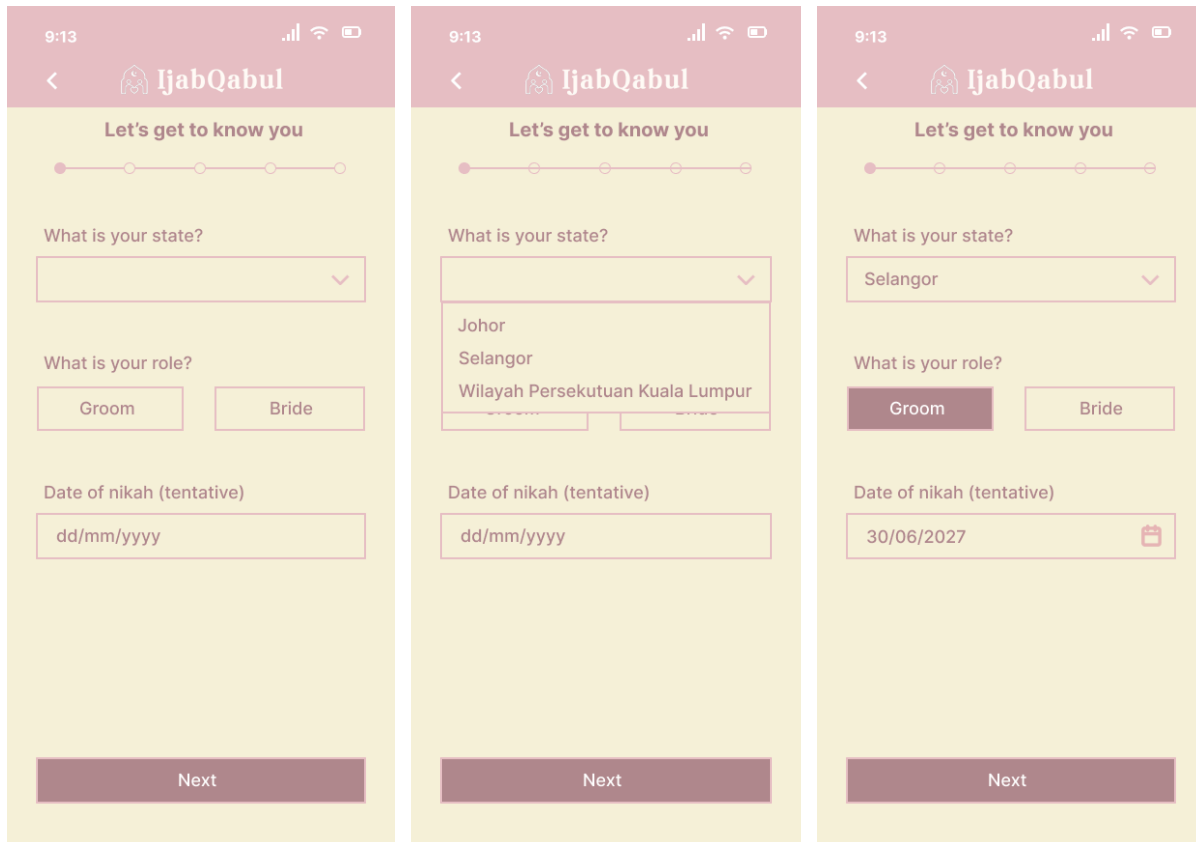


Figure 8: User Onboarding Screen

The User Onboarding Screen is designed to collect essential information that enables the application to provide a personalized wedding planning experience. Users are required to select their Malaysian state, choose their role as either a groom or bride, and enter their tentative nikah date. This information is used to generate customized wedding checklists, smart timeline recommendations, and state-specific marriage guidance. The interface adopts a clean and minimal design with clearly labelled input fields, touch-friendly buttons, and a progress indicator to ensure a simple, intuitive, and user-friendly onboarding experience on smartphone devices.

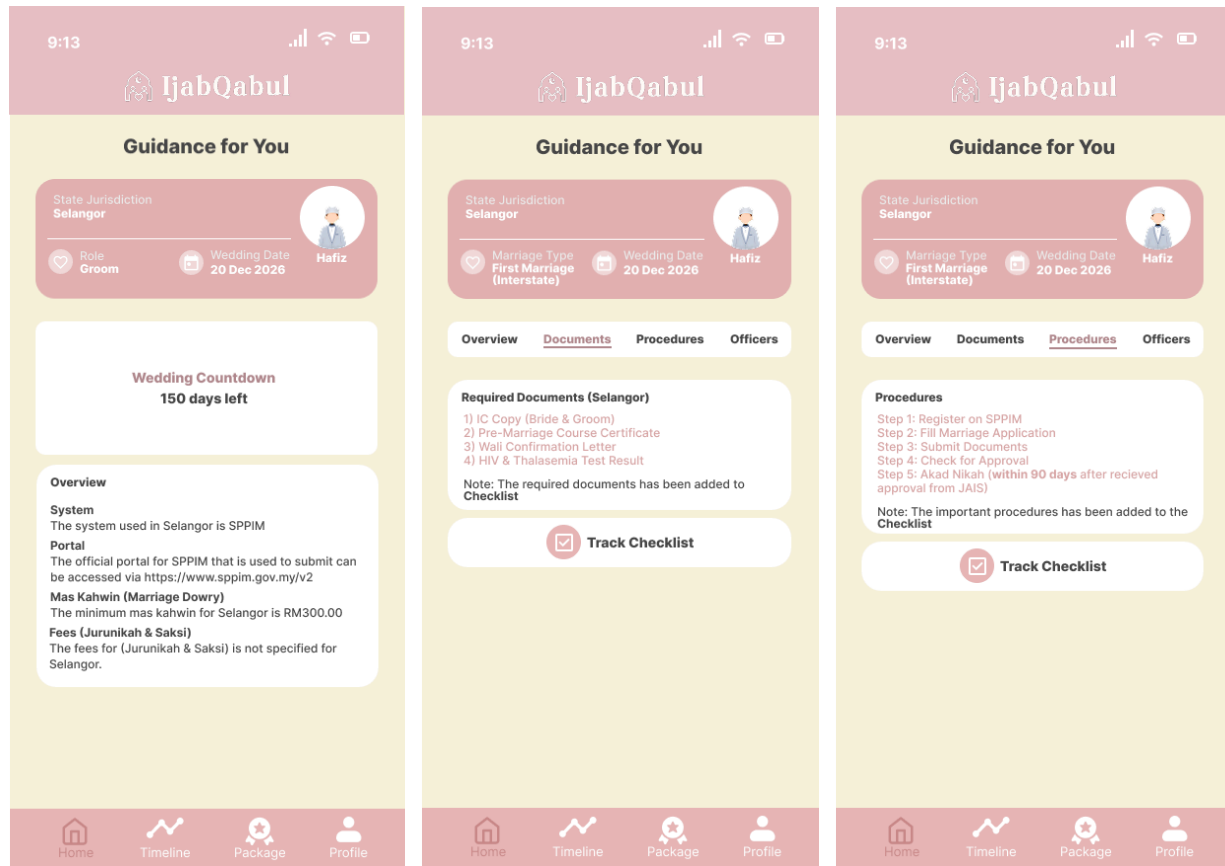


Figure 9: Home Dashboard

The Home Dashboard acts as the application's central navigation hub. It provides quick access to all major modules, including Wedding Checklist, Smart Timeline, Package Comparison, Marriage Guidance, and User Profile. Cards and icons are used to improve navigation while giving users an overview of available features.

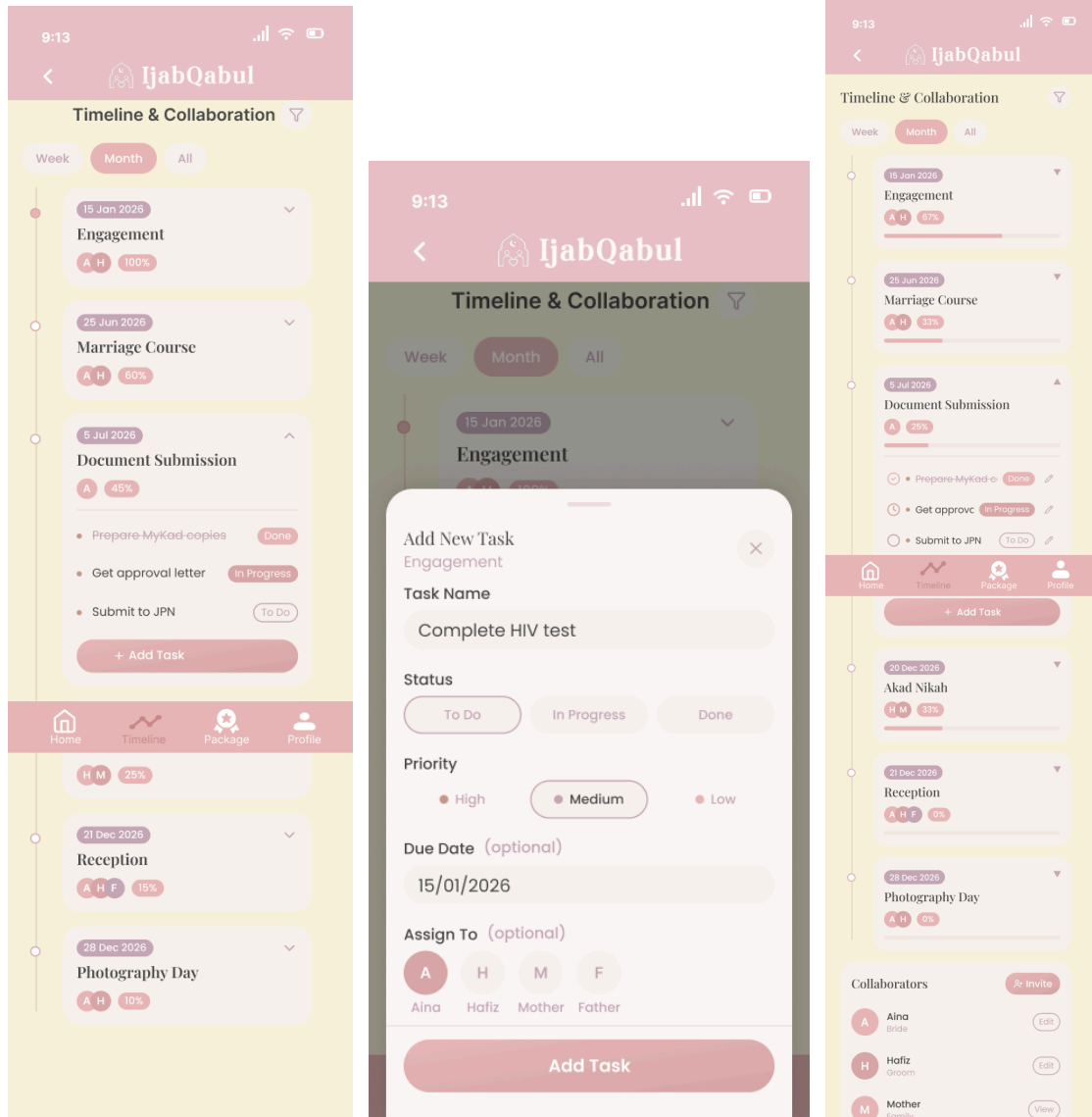
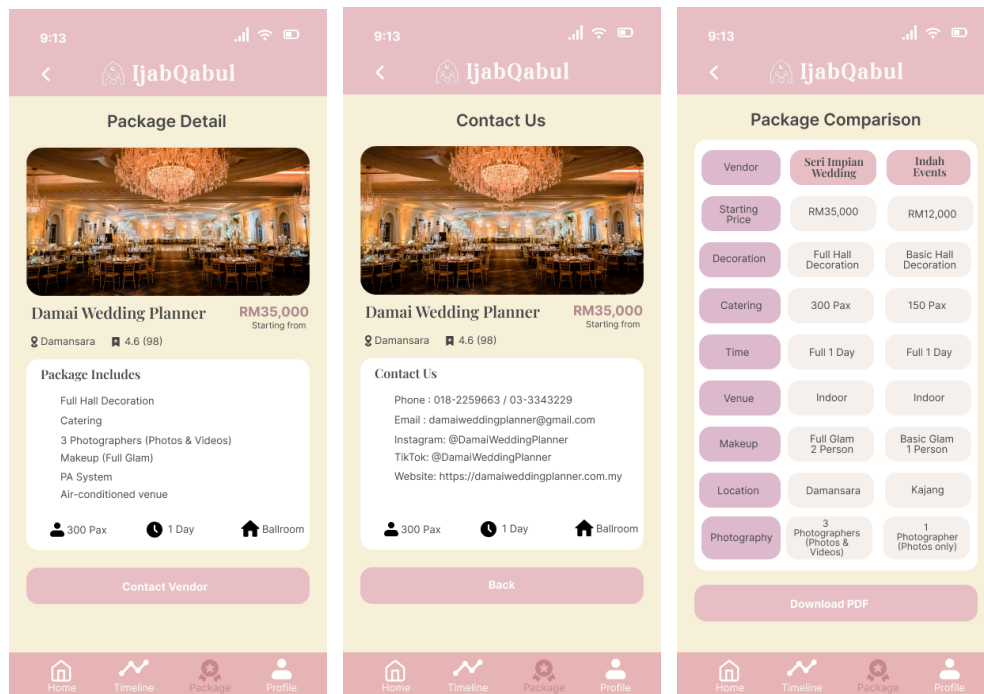
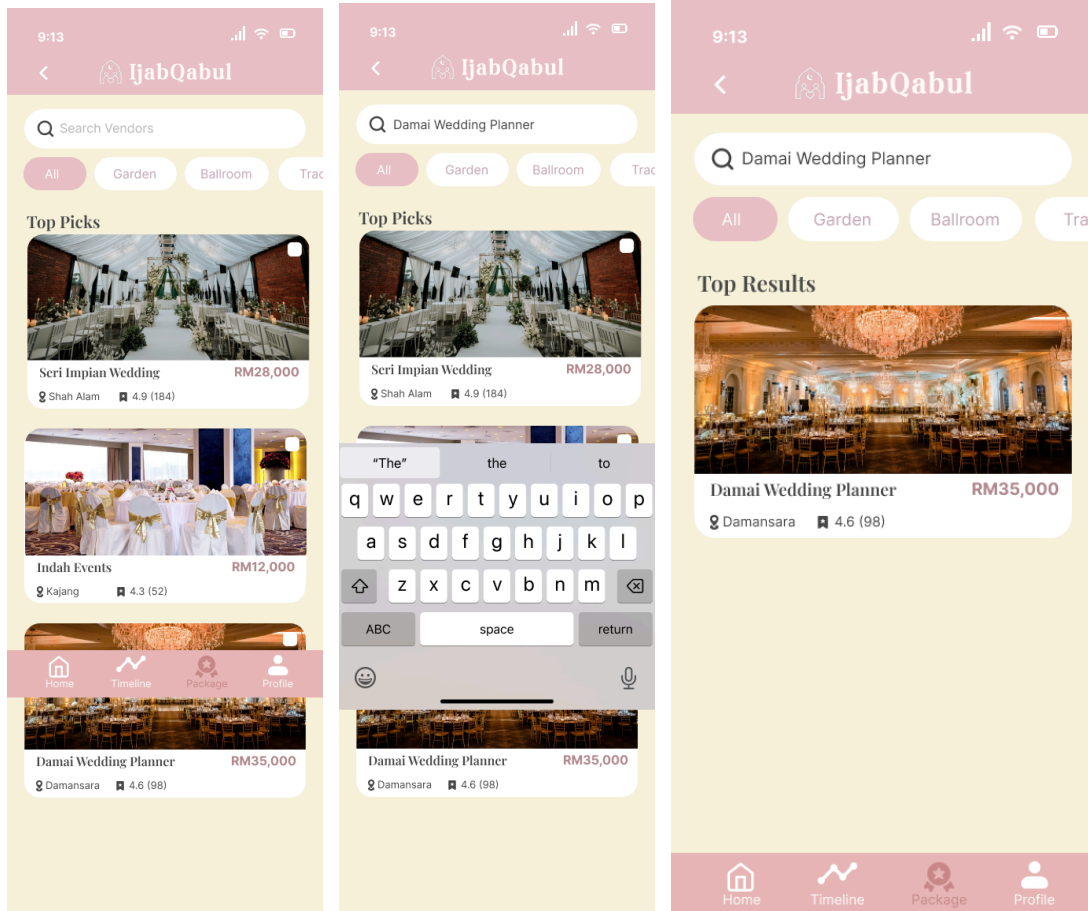


Figure 10: Personalized Wedding Checklist and Smart Timeline

The Wedding Checklist screen enables users to manage wedding preparation tasks through an organized task list. Users can add, edit, complete, or delete checklist items. Progress indicators help users monitor completed tasks and remaining activities throughout the preparation process.

The Smart Timeline screen automatically displays recommended wedding preparation milestones based on the selected wedding date. Activities are arranged chronologically, helping users complete important tasks within the recommended timeframe while reducing the need for manual scheduling.



IjabQabul		
Vendor Comparison		
	Seri Impian	Indah Events
Starting Price	RM35,000	RM12,000
Decoration	Full Hall	Basic Hall
Catering	300 Pax	150 Pax
Duration	Full Day	Full Day
Venue	Indoor	Indoor
Makeup	Full Glam	Basic Glam
Location	Damansara	Kajang
Photography	3 Photographers	1 Photographer
Generated on: 20 June 2026		
Generated by IjabQabul Wedding Planner		

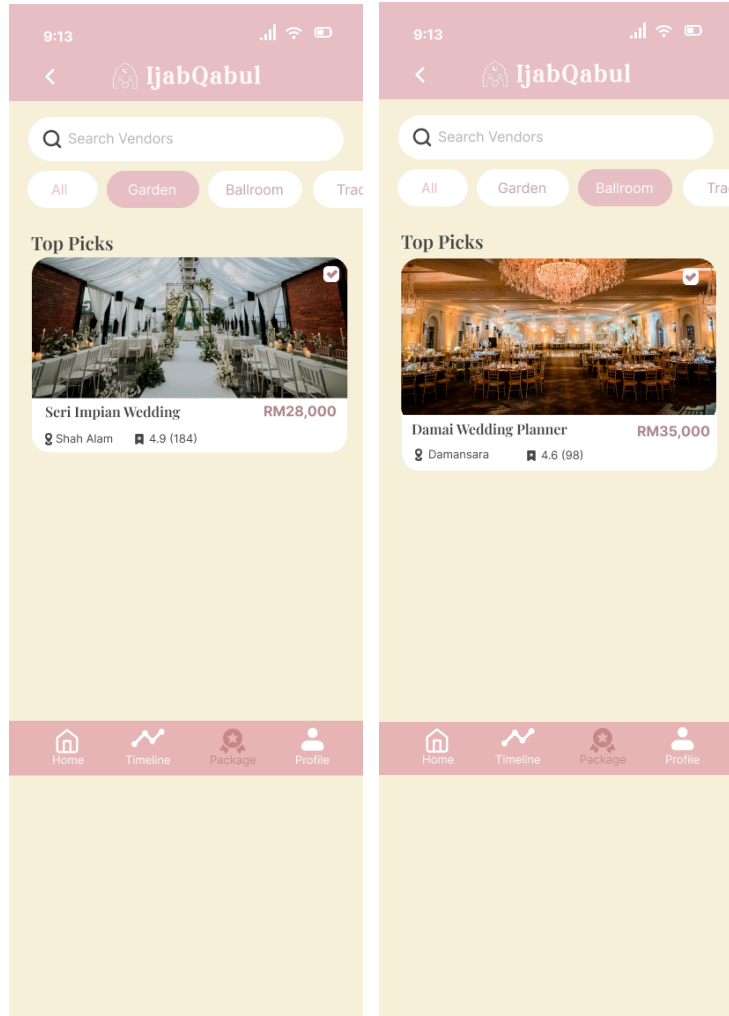


Figure 11: Wedding Package Comparison

This screen presents multiple wedding packages in a structured comparison format. Users can evaluate pricing, services, venue information, decoration themes, catering options, and customer ratings before selecting the most suitable package according to their budget and preferences.

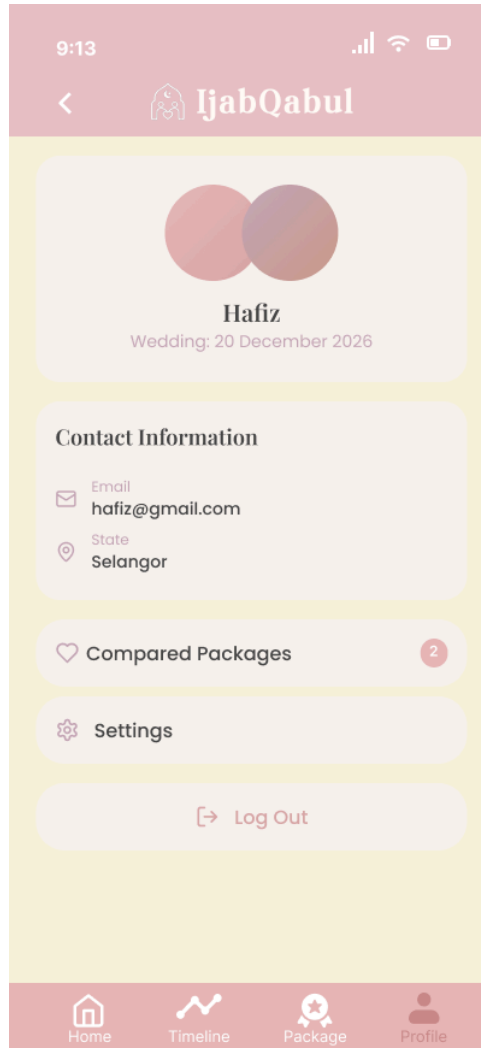


Figure 12: User Profile

The User Profile screen allows users to manage their personal information, including their name, role, selected state, and wedding date. Updates are automatically synchronized with Firebase Firestore to ensure personalized features remain accurate throughout the application.

This level of detail is typically what lecturers expect for a Project Design chapter. It explains not only what each screen contains, but also the design decisions behind it and how those decisions support usability and the project's objectives.

4. PROJECT DEVELOPMENT

The IjabQabul Wedding Planner Application is created to serve as a trusted companion for Muslim couples in Malaysia as they prepare for their wedding. At this stage, the project moves from design into development, where ideas are transformed into a real mobile app that couples can use with ease. This involves building the planned features, connecting the backend services, designing interactive screens, and ensuring that every part of the system works together in harmony. The purpose is not only to deliver a functional tool but also to create an experience that feels supportive and reliable during one of the most meaningful journeys in life.

The app is built with Flutter. Flutter allows developers to write one codebase that runs smoothly on both Android and iOS, which saves time and ensures consistency across devices. It was chosen because it provides strong performance, flexible design options through Material Design, and speeds up development with reusable widgets. On the backend, the team uses Firebase. Firebase offers services such as authentication, cloud storage, and real-time database management. By relying on Firebase, the developers avoid the complexity of setting up a traditional server, which keeps the system architecture simple while still powerful.

The development process follows a modular approach. Each feature is built as its own module before being connected to the main app. This method keeps the code organized and makes debugging easier. It also ensures that the app can be maintained more effectively in the future. Another advantage is that teamwork becomes smoother, since different developers can work on separate modules without interfering with each other's progress. For example, one team member can focus on vendor details while other works on user accounts, and both can later integrate their work into the main system without conflict.

By combining Flutter's flexibility, Firebase's reliability, and a modular development strategy, the IjabQabul Wedding Planner Application is carefully crafted to meet the real needs of couples. It is not just about technology. It is about creating an experience that feels supportive, efficient, and tailored to the cultural and religious context of its users. Couples will be able to plan their wedding with confidence, knowing that the app is designed to guide them step by step, from choosing vendors to managing packages and keeping track of important dates.

4.1 Functionality

The IjabQabul Wedding Planner Application brings together all the core features that were defined during the early stages of the project. Each function is carefully designed to respond to the real challenges faced by Muslim couples in Malaysia. Wedding preparations often involve complex tasks such as meeting legal requirements, managing budgets, and coordinating with multiple parties. By placing all of these features into one application, couples are able to plan their wedding in a way that feels more organized, efficient, and less stressful.

The system is structured into several important modules, each focusing on a specific part of the wedding planning journey. These modules include user authentication, wedding checklist management, smart timeline generation, wedding package comparison, state-specific marriage guidance, and user profile management. Each module plays its own role but they are all connected through Firebase Cloud Firestore. This ensures that user data stays consistent and can be accessed in real time across different devices, giving couples the flexibility to plan wherever they are.

By combining these modules into one system, the application does not simply provide tools. It creates a complete wedding planning experience that is tailored to the cultural and religious context of its users. Couples can trust that their information is secure, their tasks are organized, and their journey is guided step by step in a way that feels both practical and supportive.

4.1.1 User Authentication

The User Authentication module is the gateway into the IjabQabul Wedding Planner Application. Its purpose is to ensure that only authorized users can access their personalized wedding planning data. This is achieved through Firebase Authentication, which manages secure login and registration using email and password credentials.

When a new user registers, the system checks that all information is complete and properly formatted. Once the details are validated, Firebase generates a unique identifier known as a UID. This identifier becomes the main reference for all user-related data stored in the database, making it possible to link every action and record back to the correct individual.

After logging in successfully, users are directed to the main dashboard where all wedding planning features are available. The authentication state is preserved throughout the session, which means users do not need to re-enter their credentials unless they choose to log out. This creates a smooth and uninterrupted experience.

The module includes several important features:

- User registration with email validation
- Secure login and logout functionality
- Session persistence to keep users signed in
- Password strength verification for added security
- Firebase UID identification to manage user data

Together, these features make authentication reliable and user-friendly. Couples can trust that their information is protected while enjoying a seamless entry into the app's wedding planning tools.

4.1.2 Personalized Wedding Checklist

The Personalized Wedding Checklist module is designed to help couples manage every detail of their wedding preparation in an organized way. Planning a wedding can be overwhelming, and this feature reduces the risk of forgetting important tasks such as preparing documents, booking venues, arranging catering, or selecting vendors. By keeping everything in one place, couples can move through the process with greater confidence and clarity.

Each task in the checklist includes structured information such as a title, category, due date, completion status, and optional notes. The module is powered by Firebase Firestore, which supports full create, read, update, and delete operations. This means users can continuously adjust their checklist as their wedding planning evolves, adding new tasks, updating progress, or removing items that are no longer needed.

To make the experience more intuitive, completed tasks are visually distinguished from those that are still pending. This allows users to quickly see what has already been accomplished and what requires attention.

The main functionalities of this module include:

- Creating new wedding tasks to capture every important detail
- Viewing categorized checklist items for better organization
- Updating task details and tracking progress over time
- Deleting tasks that are completed or unnecessary
- Categorizing tasks into areas such as budget, venue, documents, and vendors

By offering these features, the Personalized Wedding Checklist becomes more than just a list. It acts as a guiding tool that helps couples stay on track, reduce stress, and ensure that no important step is overlooked during their wedding journey.

4.1.3 Smart Timeline Management

The Smart Timeline Management module is one of the most important features of the IjabQabul Wedding Planner Application. Unlike ordinary calendar tools that depend on manual input, this module automatically creates a structured wedding preparation timeline based on information provided by the user. Details such as the wedding date, the role of the user as bride or groom, and the Malaysian state where the marriage will take place are all taken into account.

The system then calculates recommended milestones and places them across a timeline that leads up to the wedding day. These milestones include essential activities such as attending pre-marriage courses, submitting marriage applications, booking venues, arranging catering services, and confirming photography sessions. By presenting these tasks in a clear order, couples are guided step by step through the preparation process.

This automated approach helps users avoid confusion and ensures that important tasks are completed at the right time. It reduces the need for manual planning and improves overall organization, making the wedding journey smoother and less stressful. Couples can focus on enjoying the process while trusting that the app is keeping them on track.

4.1.4 Wedding Package Comparison

The Wedding Package Comparison module is designed to help couples evaluate the different packages offered by wedding vendors. Planning a wedding often requires careful financial decisions, and this feature provides a clear comparison view that makes it easier to analyze multiple options side by side.

Each package contains important details such as price, type of venue, catering choices, decoration themes, photography services, makeup services, and guest capacity. By presenting all of this information in a consistent format, couples can quickly see the differences between packages and choose the one that best matches their budget and preferences.

The module also allows users to save selected packages for later review. This means couples can revisit their comparisons before making final decisions, ensuring that they feel confident about the choice they make.

The overall purpose of this feature is to simplify decision-making and reduce the stress that often comes with comparing multiple vendors. By organizing the information in a clear and accessible way, the application supports couples in selecting packages that truly fit their vision for the wedding day.

4.1.5 State-Specific Marriage Guidance

Marriage procedures in Malaysia are not the same across all states, and this often creates confusion for couples who are preparing for their wedding for the first time. To make the process easier, the application includes a State-Specific Marriage Guidance module that provides clear and centralized information based on the state selected by the user.

Once a state is chosen, the system presents all the relevant details in an organized way. This includes the documents that must be prepared, the step-by-step application procedures, the requirements for marriage courses, and the timelines for registration. The module also provides direct access to official Islamic Religious Department websites for each state. This ensures that couples are always guided by the most accurate and updated information available.

By offering this feature, the application saves couples from the frustration of searching across multiple external sources. It reduces the time spent gathering information and gives couples confidence that they are following the correct procedures for their state.

4.1.6 User Profile Management

The User Profile Management module gives couples a simple way to store and update their personal information within the application. This information is important because it allows the system to generate personalized recommendations such as tailored timelines and checklist suggestions that match each couple's unique situation.

Users can update details including their name, wedding date, selected state, and role as either bride or groom. Whenever changes are made, the information is automatically synchronized with Firebase Firestore. This ensures that all connected modules reflect the latest updates without requiring any extra effort from the user.

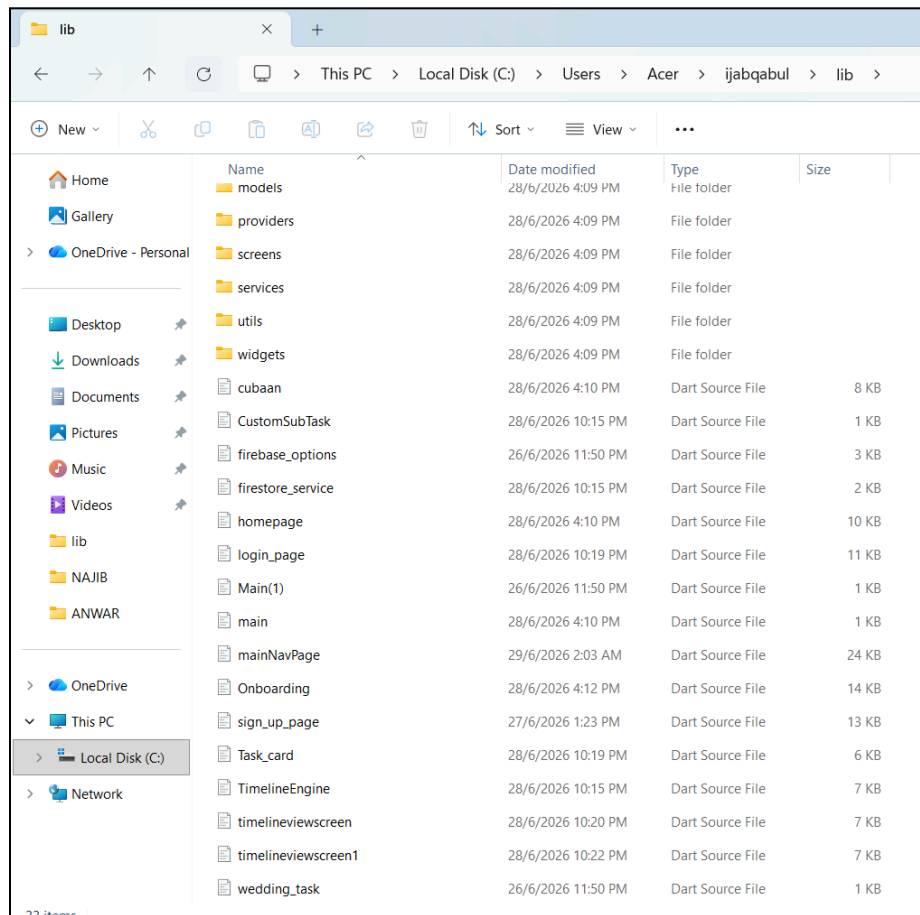
This integration creates consistency across the entire application and strengthens personalization. It means that couples receive guidance that is always aligned with their current details, making the wedding planning experience smoother and more reliable.

4.2 Code Quality

To make sure the IjabQabul Wedding Planner Application is maintainable, scalable, and easy to manage, the system is built using a modular architecture. Each feature is separated into its own component, which allows developers to work independently without interfering with other parts of the project. This structure is especially valuable in a group project environment where several members contribute at the same time.

The application follows Flutter best practices by organizing UI components, business logic, data models, and services into different directories. This separation of concerns improves readability and makes future modifications easier to implement.

The project structure is organized as follows:



Name	Date modified	Type	Size
models	28/6/2026 4:09 PM	File folder	
providers	28/6/2026 4:09 PM	File folder	
screens	28/6/2026 4:09 PM	File folder	
services	28/6/2026 4:09 PM	File folder	
utils	28/6/2026 4:09 PM	File folder	
widgets	28/6/2026 4:09 PM	File folder	
cubaan	28/6/2026 4:10 PM	Dart Source File	8 KB
CustomSubTask	28/6/2026 10:15 PM	Dart Source File	1 KB
firebase_options	26/6/2026 11:50 PM	Dart Source File	3 KB
firestore_service	28/6/2026 10:15 PM	Dart Source File	2 KB
homepage	28/6/2026 4:10 PM	Dart Source File	10 KB
login_page	28/6/2026 10:19 PM	Dart Source File	11 KB
Main(1)	26/6/2026 11:50 PM	Dart Source File	1 KB
main	28/6/2026 4:10 PM	Dart Source File	1 KB
mainNavPage	29/6/2026 2:03 AM	Dart Source File	24 KB
Onboarding	28/6/2026 4:12 PM	Dart Source File	14 KB
sign_up_page	27/6/2026 1:23 PM	Dart Source File	13 KB
Task_card	28/6/2026 10:19 PM	Dart Source File	6 KB
TimelineEngine	28/6/2026 10:15 PM	Dart Source File	7 KB
timelineviewscreen	28/6/2026 10:20 PM	Dart Source File	7 KB
timelineviewscreen1	28/6/2026 10:22 PM	Dart Source File	7 KB
wedding_task	26/6/2026 11:50 PM	Dart Source File	1 KB

Figure X: Folder Development IjabQabul

State management is handled using the Provider package. This approach separates the user interface from the business logic, which improves performance and ensures that any changes in data are automatically reflected in the interface.

Overall, this modular design strengthens maintainability, reduces code duplication, and supports parallel development among team members. It creates a clean and organized foundation that makes the application easier to expand and adapt in the future.

4.3 Packages and Plugins Implementation

The development of the IjabQabul Wedding Planner Application makes use of several Flutter packages to enhance functionality and reduce complexity during the building process. Each package is chosen with care, focusing on stability, compatibility, and strong community support so that the system remains reliable and easy to maintain.

Package	Purpose
firebase_core	Firestore initialization
cloud_firestore	Cloud database storage
firebase_auth	User authentication
provider	State management
table_calendar	Timeline and calendar display
shared_preferences	Local data storage
flutter_markdown	Display state guidance content
intl	Date and time formatting
flutter_slidable	Interactive task management
url_launcher	Open official state marriage portals

Table X: Packages

On the backend, Firebase is integrated to manage authentication and cloud storage. Firestore provides real-time database functionality, which allows user data to be instantly synchronized across devices. This ensures that couples always have access to the most updated information no matter where they are. CRUD operations are implemented across all major modules, giving users the ability to create, retrieve, update, and delete data seamlessly as they move through their wedding planning journey.

To strengthen system reliability, input validation and error handling are applied throughout the application. Every user input is checked before being processed. For example, email addresses must follow the correct format, passwords must meet security requirements, required fields cannot be left empty, and wedding dates cannot be set in the past. In addition, Firebase exceptions such as network errors or authentication failures are handled gracefully.

Instead of technical error codes, users receive clear and friendly messages that guide them on what to do next.

This careful integration of Flutter packages and Firebase services ensures that the application is not only functional but also dependable. Couples can trust that their data is secure, their inputs are validated, and any issues are managed in a way that keeps the experience smooth and supportive.

4.3 Collaborative Tool

The IjabQabul Wedding Planner Application is developed through teamwork, and GitHub serves as the main platform for collaboration. It manages the source code and coordinates contributions from different developers, allowing everyone to work on separate modules at the same time without creating conflicts in the main project.

A clear branching strategy is followed to keep the workflow organized. The main branch contains the stable version of the application, while the develop branch is used to integrate features that have been completed. Each module is built in its own branch, such as authentication, checklist, or timeline, before being merged into the larger system.

Before any code is merged, it goes through a review process. This ensures that the work is consistent, functional, and aligned with coding standards. By reviewing code carefully, the team reduces errors and makes sure that only tested and stable features are added to the project.

GitHub also acts as a central hub for documentation, progress tracking, and version control. Weekly updates are recorded, and contributions from each team member are documented to maintain transparency and accountability throughout the development process.

This collaborative approach makes teamwork smoother, keeps the project organized, and ensures that the application grows in a reliable and structured way.

CONCLUSION

The development of the IjabQabul Wedding Planner Application shows how modern mobile technologies can be applied to solve real challenges faced by Muslim couples in Malaysia during wedding preparation. Planning a wedding is often complex and stressful, involving tasks such as budgeting, coordinating with vendors, preparing legal documents, and following state-specific marriage procedures. This project addresses those challenges by bringing all essential tools together into one centralized mobile application.

The application was built using Flutter for the frontend and Firebase for the backend. Flutter makes it possible to create a responsive and visually consistent interface across both Android and iOS platforms. Firebase ensures secure authentication, real-time database management, and reliable cloud storage. Together, these technologies reduce system complexity while improving scalability and performance.

A modular architecture was adopted throughout development. Features such as user authentication, wedding checklist, smart timeline, package comparison, and marriage guidance were built and tested independently before being integrated into the main system. This approach improved maintainability and supported effective collaboration among team members, who used GitHub for version control and task management.

In terms of functionality, the application achieves its objectives by giving couples a structured and organized way to manage their wedding preparations. The personalized checklist helps track tasks efficiently, while the smart timeline automatically generates milestones based on user inputs. The package comparison and state-specific guidance modules assist couples in making informed decisions, ensuring that financial planning and legal requirements are properly addressed.

There is still room for future improvement. Advanced features such as AI-based budget recommendations, vendor booking integration, notification reminders, and real-time chat collaboration could be introduced to enhance usability and provide a more complete wedding planning experience.

In conclusion, the IjabQabul Wedding Planner Application successfully meets its intended goals by offering a practical, user-friendly, and centralized solution for wedding planning. The project demonstrates how mobile application technologies can simplify complex real-world processes while improving user experience, efficiency, and organization.

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